

FIDES Workshop



[Safety and Reliability Training](#)

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ANZEN

ANZEN PROPRIETARY INFORMATION





Workshop Contents

- Reliability estimation methods
- Why Fides?
- How is Fides implemented
- Physical Factor
- Part Manufacturing Factor
- Process Factor

Reliability Estimation Methods

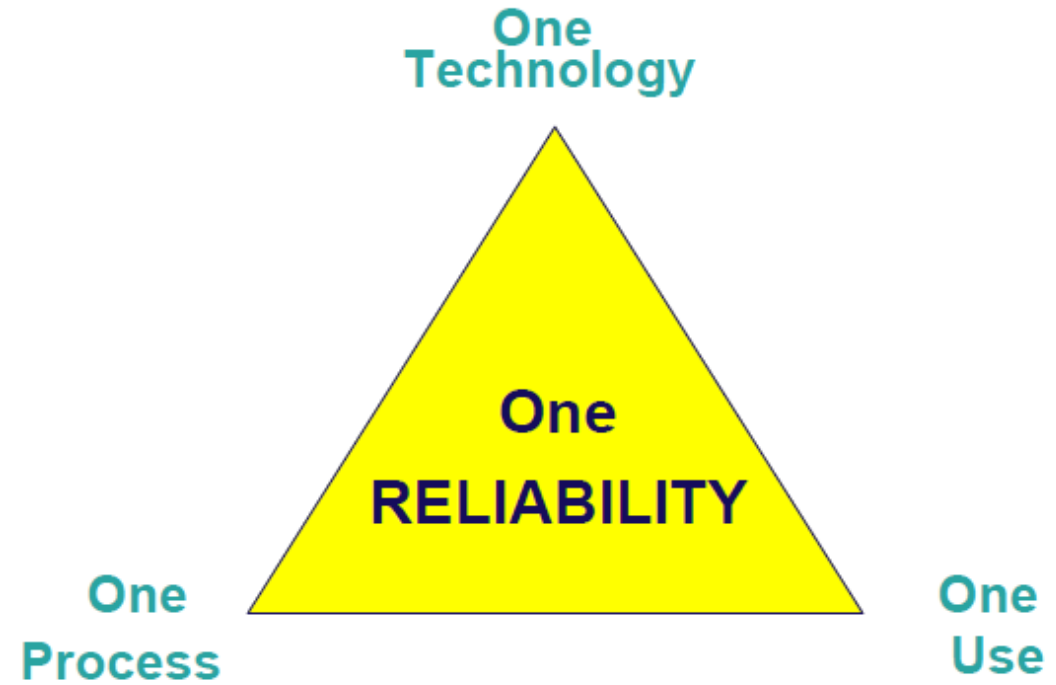
- Data from the supplier
- Internal data based on similar components
- Highly Accelerated Life Test (HALT)
- Standards:
 - MIL-HDBK-217F
 - Fides
 - ...

Why Fides?

- Reliability calculation for electronic components and systems
- Developed in 2004 by Airbus, Thales and other aeronautical companies
- Based on the physics of failures
- Realistic assessment

How is it implemented?

- Technology: Components
- Use: Mission profile
- Process: Life cycle



$$\lambda = \lambda_{\text{Physical}} \cdot \prod_{\text{PM}} \cdot \prod_{\text{Process}}$$

FIDES formula (I)

- The main FIDES formulae are of the form:

$$\lambda = \lambda_{\text{physical}} * \pi_{\text{Part_manufacturing}} * \pi_{\text{Process}}$$

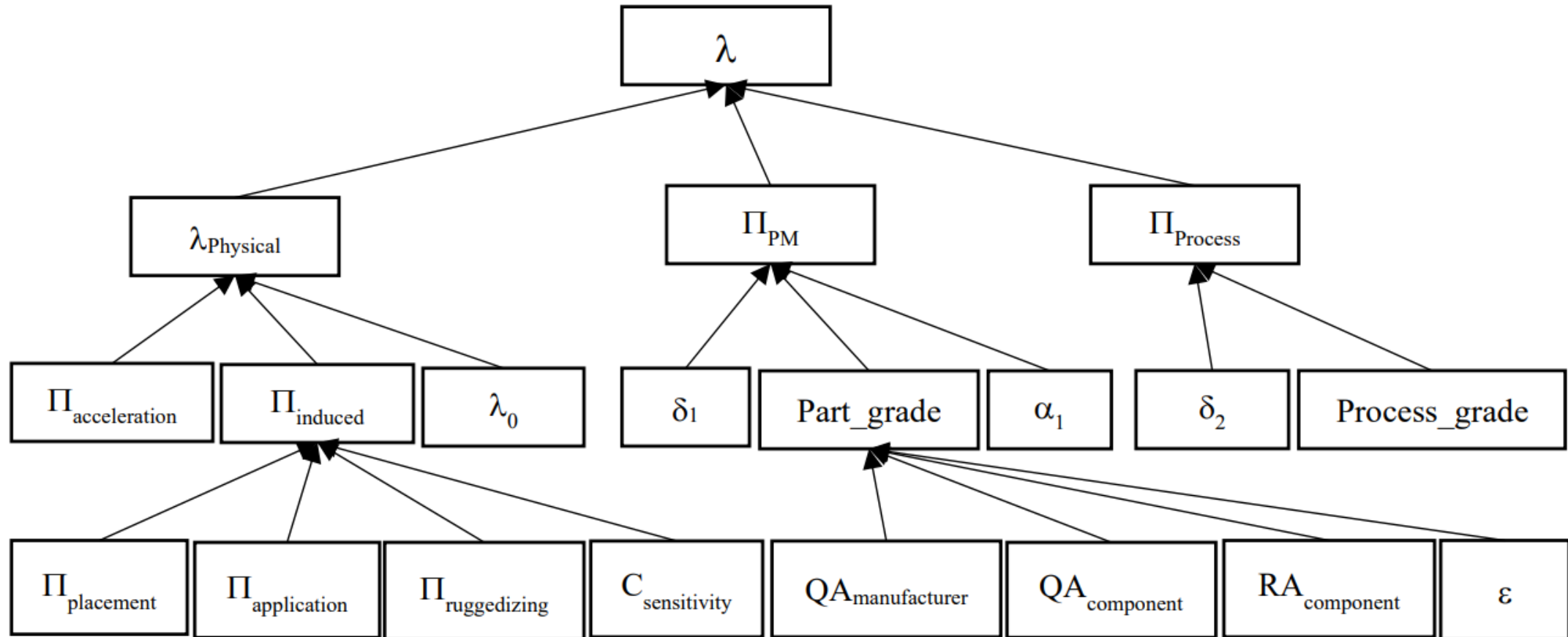
Physical contribution *Quality and level of qualification of the part* *Quality of the process at equipment & system levels*

with

$$\lambda_{\text{physical}} = \lambda_{0_part} * \sum \pi_{\text{acceleration}} * \pi_{\text{Induced}}$$

Impact of thermal cycling, electrical, thermal, chemical, mechanical & humidity influences *Impact of application on the part (placement, application & ruggedizing)*

FIDES formula (II)



Physical Contribution: Mission profile + overstress

$\pi_{acceleration}$

Thermal-Electrical

Temperature Cycle

Mechanical

Humidity

Chemical

$\pi_{induced}$

Placement

Application

Ruggedising

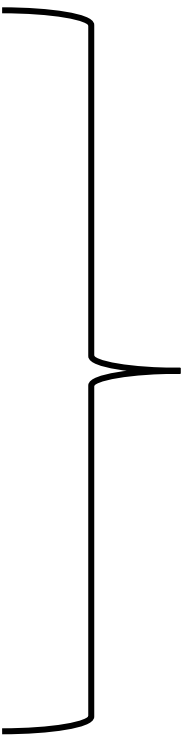
Component sensitivity

$$\lambda_{\text{Physical}} = \left[\sum_{\text{Physical_Contributions}} (\lambda_0 \cdot \Pi_{\text{acceleration}}) \right] \cdot \Pi_{\text{induced}}$$

Acceleration factor (mission profile)

What do we need?

- Definition of phases and cycles
- State: On / Off / Stand by
- Temperature: average, max, delta
- Relative Humidity
- Mechanical: Random Vibrations
- Chemical:
 - Saline
 - Application: Close to the motor?
 - Environmental
 - Protection level: Hermetic?



For each phase

Induced Factor (I): Placement

Type of function	Placement Factor
Digital non-interface function	1.0
Digital interface function	1.6
Analogue low level (<1A) non-interface function	1.3
Analogue low level (<1A) interface function	2.0
Analogue power (>1A) non-interface function	1.6
Analogue power (>1A) interface function	2.5

Can be determined at component or board level

Induced Factor (II): Application

¿Favorable, moderate or unfavorable situation?

- User type for each phase
- User qualification level for each phase
- System mobility
- Product manipulation
- Type of electrical network
- Exposure to human activity
- Exposure to machine disturbances
- Exposure to weather

Induced Factor (III): Application

- User type: Represents professionalism, respect of procedures, influence of operational stresses.
- User qualification: Represents the control level of the user or the operator in an operational context.
- System mobility: Represents problems related to the possibilities of moving the system.
- Product manipulation: Represents the risks of false manipulations, shocks, falls, etc.
- Power supply type: Refers to the level of electrical disturbance expected on power supplies and signals: power on, power supply switching, connection/disconnection.
- Exposure to human activity: Represents exposure to problems related to human activity: shock, change in the final use, etc.
- Exposure to machine disturbances: Represents problems related to functioning of machines, motors, actuators such as shocks, overheating, electrical disturbances, aggressive pollutants.
- Exposure to the weather: Represents exposure to rain, hail, frost, sandstorm, lightning, dust, etc.

Induced Factor (IV): Application

The main rule to be respected for answering these questions is to reply to the question with respect to the appropriate level, i.e. product level or system level..

The preferred point of view is shown in the following table.

Criterion	Level
User type in the phase considered	Complete system
User qualification level in the phase considered	Product user in the complete system
System mobility	Complete system
Product handling	Product
Type of electrical network for the system	System and product

Criterion	Level
Product exposure to human activity	Product in the complete system
Product exposure to machine disturbances	Product in the complete system
Product exposure to the weather	Product in the complete system

Induced Factor (V): Ruggedising

- Reliability recommendations which satisfaction level should be evaluated

Recommendation	Satisfaction level	Weight
Make an analysis of failure cases that could lead to a failure propagation	N2=partially applied	4
Carry out a process analysis of implementation and maintenance operations	N4=fully applied and described in a procedure	4
...	...	...

Default Value = 1.7

Induced Factor (VI): Component Sensitivity

- It is calculated separately for each type of component
- It varies from 1 to 10
- It depends on: Electrical OverStress, Thermal OverStress and Mechanical OverStress
- Example for resistors:

	Relative sensitivity (mark out of 10)			C _{sensitivity}
	EOS	MOS	TOS	
"Minimelf" high stability (RS) common (RC) low power film resistor	5	2	4	3.85
Power film resistor	2	3	1	2.25
Low power wirewound accuracy resistor	2	1	3	1.75
Power wirewound resistor	2	3	1	2.25
Trimming potentiometer (CERMET)	1	5	2	2.50
Resistive chip	5	4	6	4.75
SMD resistive network	4	5	3	4.25
High stability bulk metal foil accuracy resistor	6	6	4	5.8

Part Manufacturing Factor (I): General definition

$$\Pi_{PM} = e^{1.39 \times (1 - \text{Part_Grade}) - 0.69}$$

For active components:

$$\text{Part_Grade} = \left\lceil \frac{(QA_{\text{manufacturer}} + QA_{\text{component}} + RA_{\text{component}}) \times \varepsilon}{36} \right\rceil$$

For passive components:

$$\text{Part_Grade} = \left\lceil \frac{(QA_{\text{manufacturer}} + QA_{\text{component}}) \times \varepsilon}{24} \right\rceil$$

$$0,5 \leq \pi_{PM} \leq 2$$

Default Value = 1,6 for passive / 1,7 for active

Process Factor: Quality of the life cycle process

Each phase has a relative weight

Phase	Contribution
Specification	8 %
Design	16 %
Manufacturing	20 %
Integration into equipment	10 %
Integration into system	10 %
Operation and Maintenance	18 %
Support Activities	18 %

- Recommendations are defined for every phase
- Each recommendation has a weight
- Their level of satisfaction shall be evaluated
 - N1: Not applied
 - N2: Partially applied
 - N3: Practically applied
 - N3: Fully applied

$$1 \leq \pi_{process} \leq 8; \text{ Default Value} = 4$$

MIL-HDBK-217F vs Fides

MIL-HDBK-217F	Fides
Conservative	Realistic
Based on statistical (outdated) data	Based on the physics of components
Only operational	Operational and off
Simple environment	Complex environment
Considers processess	Not considers processess

Any questions?



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